

BRENNER FIELD, NE, USAWMO: **725533**Lat: **40.080N**Lon: **95.592W**Elev: **299**StdP: **97.79**Time Zone: **-6.00 (NAC)**Period: **00-19**WBAN: **94957****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-15.6	-22.2	0.5	-17.3	-20.1	0.7	-14.6	11.9	1.4	10.9	-0.2	3.0	320	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	35.2	25.1	33.5	24.5	32.1	23.9	26.7	32.5	25.8	31.5	25.0	30.5	5.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	21.0	30.4	24.1	19.7	29.5	23.2	18.6	28.5	85.4	32.6	81.3	31.6	77.4	30.3	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.3	8.4	-22.3	38.1	2.6	1.7	-24.2	39.4	-25.7	40.4	-27.2	41.4	-29.1	42.6		
(5)			-22.6	28.4	2.6	0.7	-24.4	29.0	-25.9	29.4	-27.4	29.8	-29.2	30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	-2.7	-1.3	6.1	12.4	18.1	23.4	25.4	24.0	19.9	12.8	5.6	-0.9
	DBStd	11.27	6.69	6.76	6.60	5.55	4.70	3.33	3.15	3.14	4.48	5.33	5.83	6.23
	HDD10.0	1439	394	321	157	38	2	0	0	0	1	32	154	341
	HDD18.3	3041	652	551	382	189	65	3	1	1	34	184	382	597
	CDD10.0	2153	1	3	36	109	252	402	476	433	299	119	22	2
	CDD18.3	713	0	0	2	11	56	156	218	176	82	13	0	0
	CDH23.3	7348	0	1	31	142	554	1586	2357	1713	822	141	4	0
	CDH26.7	2984	0	0	4	35	174	637	1073	721	305	34	0	0
Wind	WSAvg	3.9	4.0	4.1	4.6	4.8	4.3	3.8	3.0	2.8	3.3	3.8	4.1	4.0
Precipitation	PrecAvg	834	17	27	48	81	117	127	124	89	74	63	46	26
	PrecMax	1374	40	79	139	178	266	237	577	212	182	136	107	83
	PrecMin	546	1	0	1	22	21	29	32	19	2	5	0	0
	PrecStd	187	9	19	35	32	60	54	117	48	50	37	31	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.7	21.0	26.6	30.4	33.2	35.5	37.6	37.1	34.8	30.2	23.6	17.8
		MCWB	8.7	11.6	16.2	18.8	21.5	24.4	24.7	24.3	23.1	20.6	15.1	13.1
	2%	DB	12.8	15.5	23.3	27.0	30.5	33.6	35.5	34.3	32.3	27.0	20.6	13.5
		MCWB	6.8	9.2	14.4	17.0	20.7	24.0	25.5	25.1	23.0	18.1	13.9	8.8
	5%	DB	9.8	11.8	20.0	24.5	28.5	32.2	33.8	32.5	30.3	24.4	17.6	10.7
		MCWB	5.2	6.9	13.3	16.1	19.8	23.4	25.3	24.6	22.1	17.3	12.2	6.8
	10%	DB	6.6	8.7	16.9	22.1	26.6	30.4	32.1	30.6	28.0	22.0	15.1	8.1
		MCWB	3.4	4.9	11.7	15.2	19.2	22.8	24.8	23.6	20.5	16.1	10.6	5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	12.9	18.1	20.2	23.5	26.7	28.3	27.9	25.2	22.5	16.8	14.3
		MCDB	12.6	17.2	23.7	27.4	29.6	32.9	33.1	33.3	31.8	27.8	20.9	16.3
	2%	WB	7.3	9.8	16.3	18.8	22.2	25.4	27.0	26.4	24.1	20.6	15.0	9.8
		MCDB	12.1	15.1	20.6	24.6	28.3	31.5	32.5	32.0	30.1	24.5	18.7	12.6
	5%	WB	5.5	7.2	14.5	17.3	21.1	24.4	26.2	25.5	23.0	18.5	13.0	7.0
		MCDB	9.7	11.7	19.3	23.1	27.0	30.3	31.9	30.9	28.4	22.5	17.1	10.2
	10%	WB	3.4	5.0	11.6	15.6	20.0	23.5	25.2	24.5	22.0	16.5	10.9	5.1
		MCDB	6.4	8.3	17.1	21.2	25.2	29.1	30.9	29.2	26.7	21.0	14.9	8.2
Mean Daily Temperature Range	5% DB	MDBR	10.8	10.9	12.1	12.5	11.6	11.3	11.2	11.5	12.9	12.5	11.8	10.2
		MCDBR	15.8	16.3	16.8	16.1	13.9	12.7	12.8	13.7	13.9	15.5	15.6	14.5
		MCWBR	10.8	10.7	9.6	8.8	6.4	5.5	5.6	6.2	6.1	8.1	9.4	10.2
		MCDBR	14.8	15.3	13.6	13.8	12.0	11.3	11.3	11.4	12.3	12.2	13.5	13.2
	5% WB	MCWBR	10.5	10.7	8.2	8.5	6.1	5.9	5.9	6.1	6.1	7.8	9.1	9.9
Clear-Sky Solar Irradiance	taub		0.280	0.304	0.329	0.380	0.411	0.420	0.425	0.424	0.379	0.335	0.299	0.284
	taud		2.483	2.428	2.419	2.294	2.273	2.287	2.273	2.293	2.371	2.475	2.515	2.482
	Ebn at Noon		896	917	927	894	869	858	850	842	861	869	866	860
	Edh at Noon		76	93	105	127	133	131	132	126	109	88	74	71
All-Sky Solar Radiation	RadAvg		2.15	3.01	3.91	4.95	5.80	6.49	6.47	5.71	4.74	3.37	2.37	1.83
	RadStd		0.17	0.24	0.34	0.31	0.49	0.47	0.45	0.28	0.32	0.37	0.19	0.18